

Rapid Microwave Sintering of Silver Nanoparticles

Rapid electrical sintering using microwave power at about 1.8 GHz, without physically contacting the nanoparticle layer, is the most promising sintering method. It is non-destructive, energy-efficient and fast. Also, it enables real-time monitoring. Efficient sintering is obtained by focusing the electric field on the PCB. Here is how it is done

RATHINDRA NATH BISWAS

Printed circuit boards (PCBs) are used in almost all electronic devices today. Metallic nanoparticle inks and pastes are used to print high-quality conductors on low-cost flexible substrates using methods like inkjet or gravure. Silver nanoparticles are sintered by heating the printed structure when the printed pattern attains near-bulk metal conductivity.

Oven or hot-plate curing at above 200°C for more than 10 minutes is usually required for efficient sintering. However, curing at such high temperature for a prolonged period is challenging, especially when processing on a flexible substrate. To overcome this problem, alternative sintering methods such as laser sintering, microwave sintering, pulsed light sintering and contactless rapid electrical sintering (RES) have been developed.

Of all these methods, RES using microwave power at about 1.8 GHz, without physically contacting the nanoparticle layer, is the most promising. It is non-destructive, energy-efficient and fast as the conductivity of patterned particles increases by several orders of magnitudes in just a few microseconds. Also, it enables real-time monitoring and provides excellent quality. Efficient sintering is obtained by focusing the electric field on the PCB.

DC RES

Fig. 1 shows a DC RES setup, where sintering electrodes are in contact with

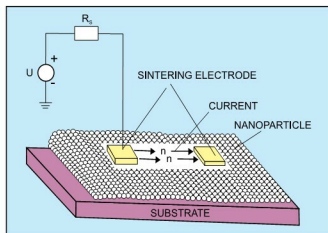


Fig. 1: DC RES setup

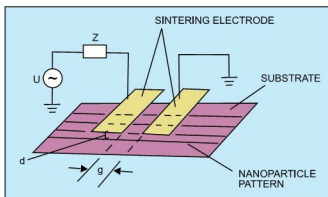


Fig. 2: AC RES setup

the nanoparticle layer. When a DC voltage (U) is coupled between the sintering electrodes, a current flows to cause local heating in the layer. The applied voltage generates power dissipation in the nanoparticle layer in the region between the sintering electrodes. This initiates sintering process and the structure undergoes a rapid transition in conductivity. Series resistor R_s limits the maximum current once the structure is sintered.

This process has several disadvantages.

When a large contact resistance is present between the electrode and the nanoparticle material, it limits processing speeds and sometimes sintering becomes impossible. Further, it has been found that sintering large areas with constant DC voltage leads to the formation of a sintered narrow filament from one electrode to the other, while leaving the remaining area unsintered. This is because the electrical current cannot be directed to unsintered areas once a conducting route has been formed.

AC RES

The problem faced with DC RES can be overcome by using a contactless sintering head with an AC source suitably designed to have adequate biasing so that the sintering power is evenly distributed below the electrodes. The AC source enables RES without physically contacting the printed layer. Due to the high initial resistivity of the nanoparticle layer, which is typically more than 0.01 ohm.m, efficient power delivery to the layer is best obtained when coupling via the near-field electric field of the sintering head.

When the adjacent electrodes are